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Santa Clara University	ELEN 115 Spring 2024	Dr. Shoba Krishnan
Laboratory #2: Single time constant circuits		

I. Objectives

- To construct a simple RC single time constant circuit and do transient analysis in LTSpice.
- To understand frequency behavior and do ac analysis for simple RC circuits.

II. Laboratory Procedure:

Part 1: Schematic Capture

For this part of the lab, please create the single time constant circuit shown in Figure 1 in LTSpice.

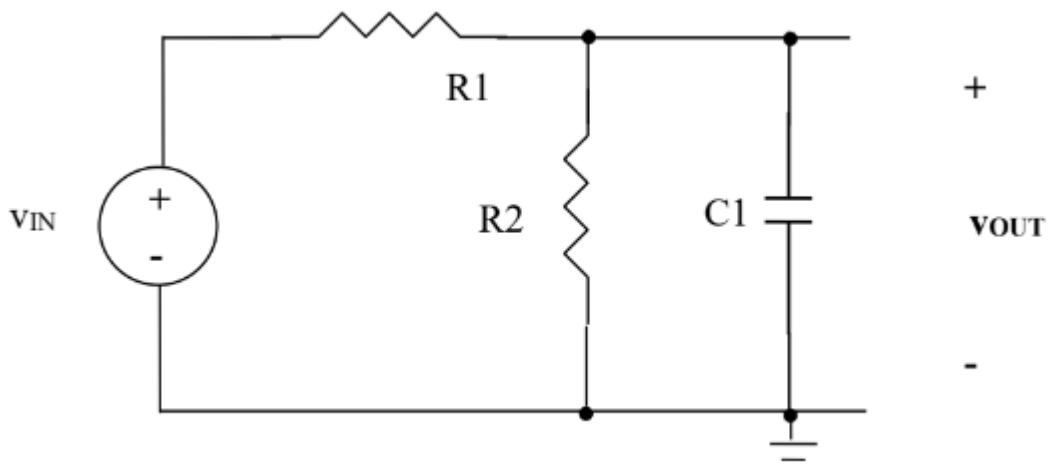


Figure 1. Single time constant circuit

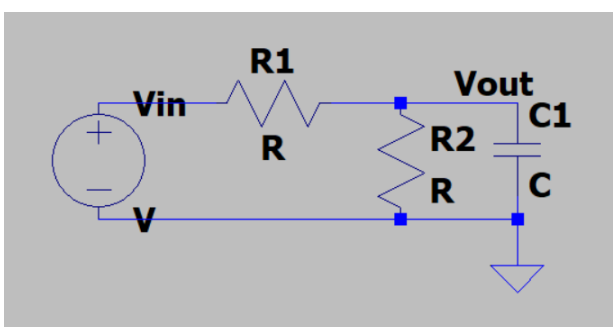


Figure made in LTSPICE

Part 2: Transient Analysis- Pulse Input

(1) Assign component value for R1 = 20kΩ, R2 = 20kΩ and C1 = 0.1μF.

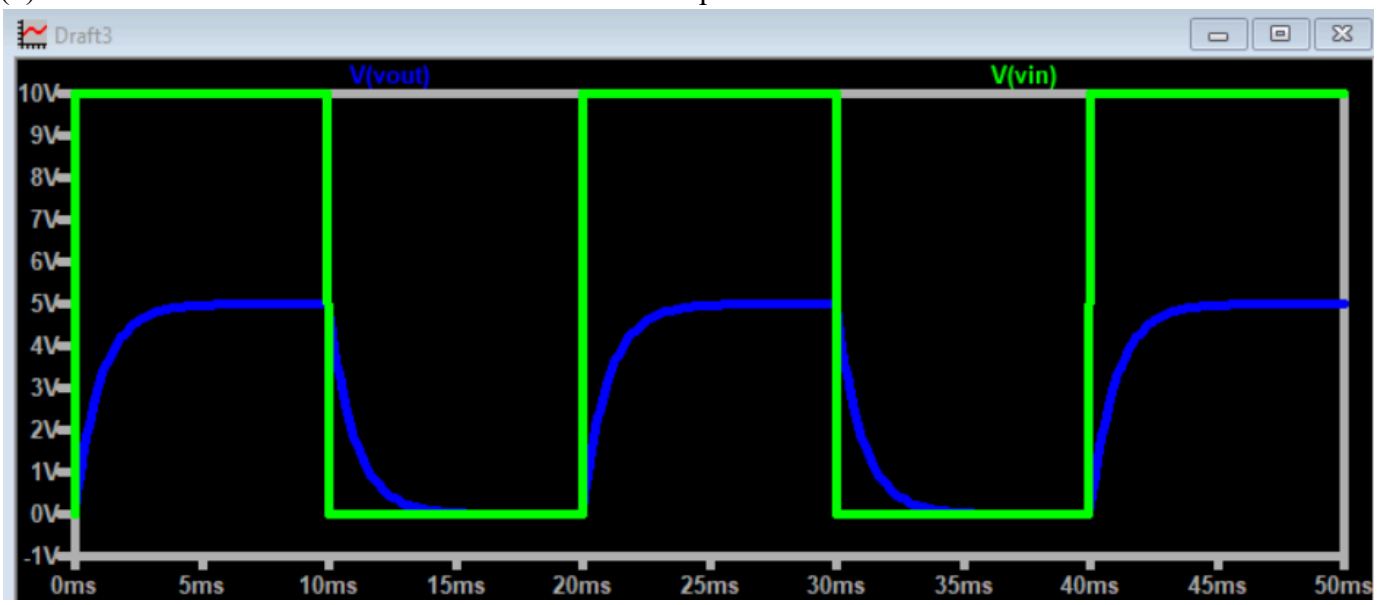
(2) Make VIN a pulse input that goes from 0V to 10V and has a pulse width of 10msec and period of

20msec. The rise and fall times should both be 1nsec. Set VIN pulse delay time = 0. (3) Set the initial condition on the capacitor to be 0V.

To do this, make a SPICE directive with the following text: ".ic V(Vout)=0" where Vout is the name of the output node.

(4) Run a Transient Analysis for 50ms, using a 10us time step.

(5) Plot VIN and VOUT as a function of time on the same plot.

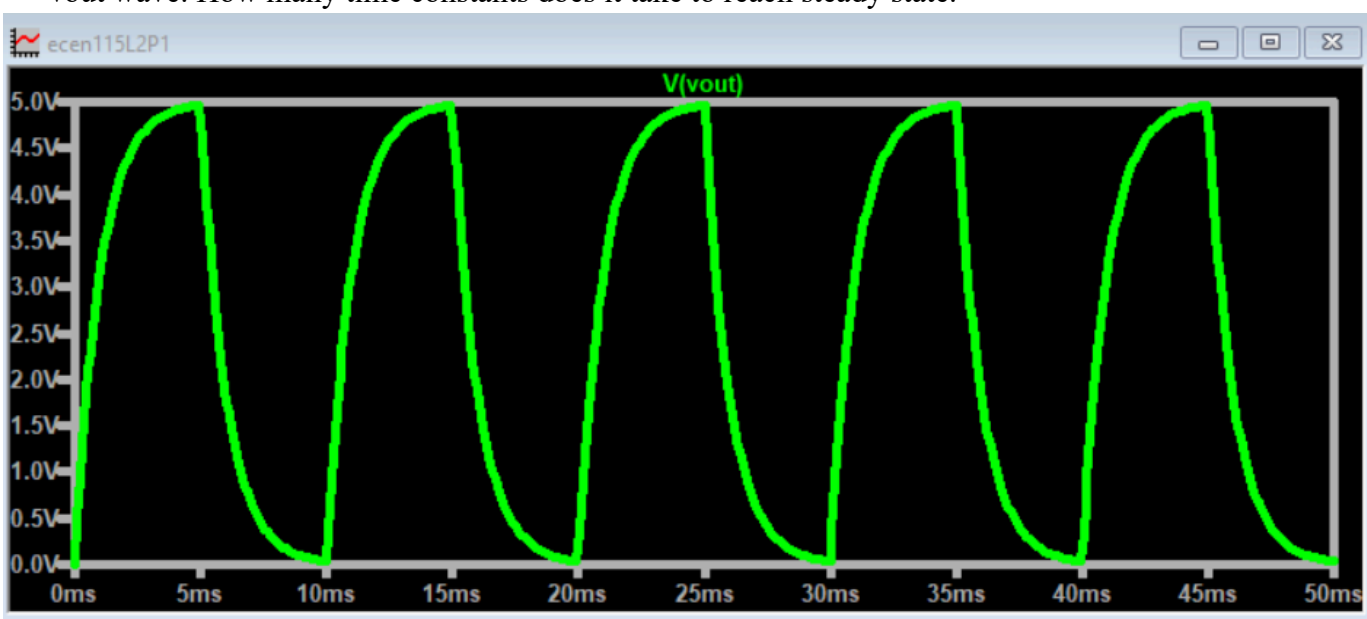


Do the results look as expected? Why or why not?

Yes they are as expected. The voltage division of the two resistors is why the steady state is at 5V and not 10, and because of capacitor behavior, it starts like a sc where no voltage flows and overtime turns into an open circuit. This results in an initial/transient state of no voltage that increases to 5 and at steady state creates a voltage divider circuit (5V).

Make sure to take a screen capture of the simulation waveforms.

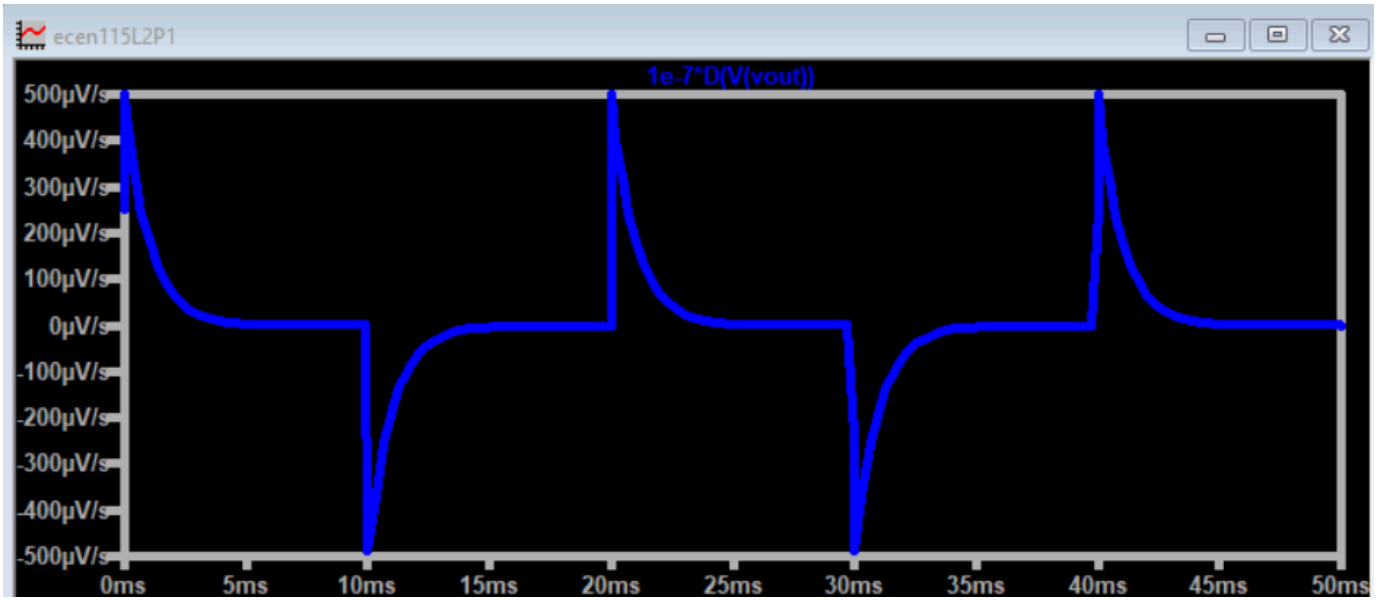
(6) Change the VIN pulse width and period (and transient runtime) to produce a full charge/discharge Vout wave. How many time constants does it take to reach steady state.



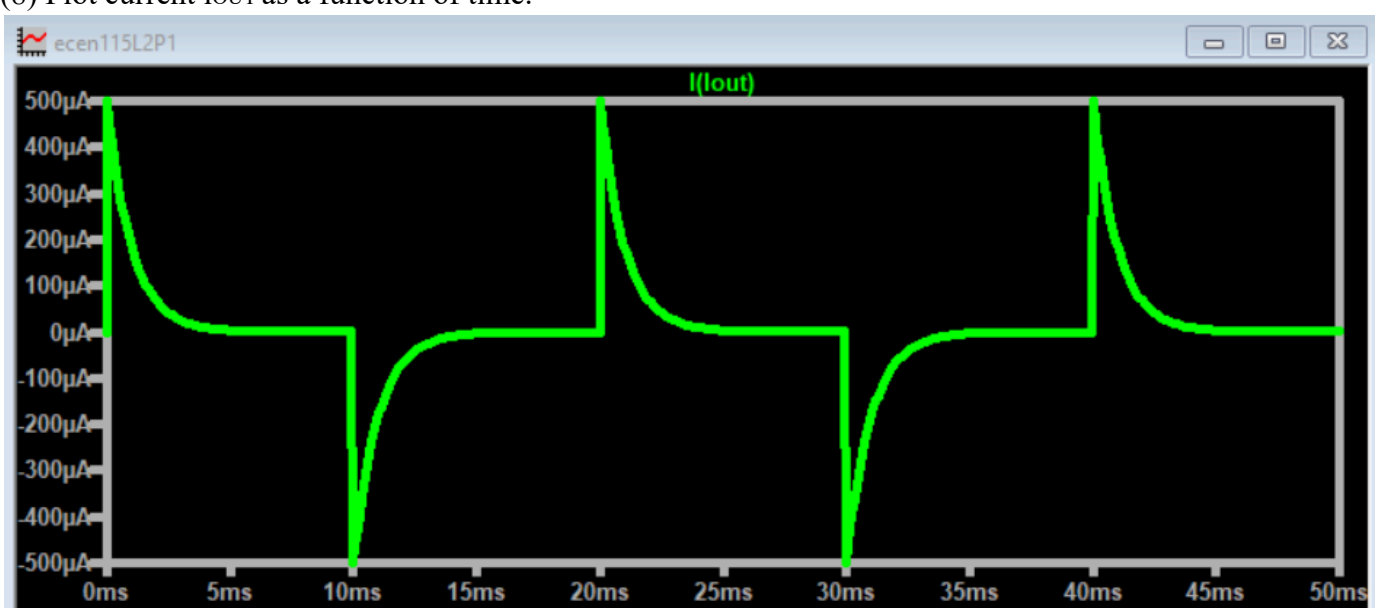
It takes five time constants. Steady state happens at 5ms and one time constant is

1/(10000*.1*10^-6)=.001s. 5ms/1ms=5

(7) Plot C*dVout/dt (use: 1e-7*D(V(Vout)).



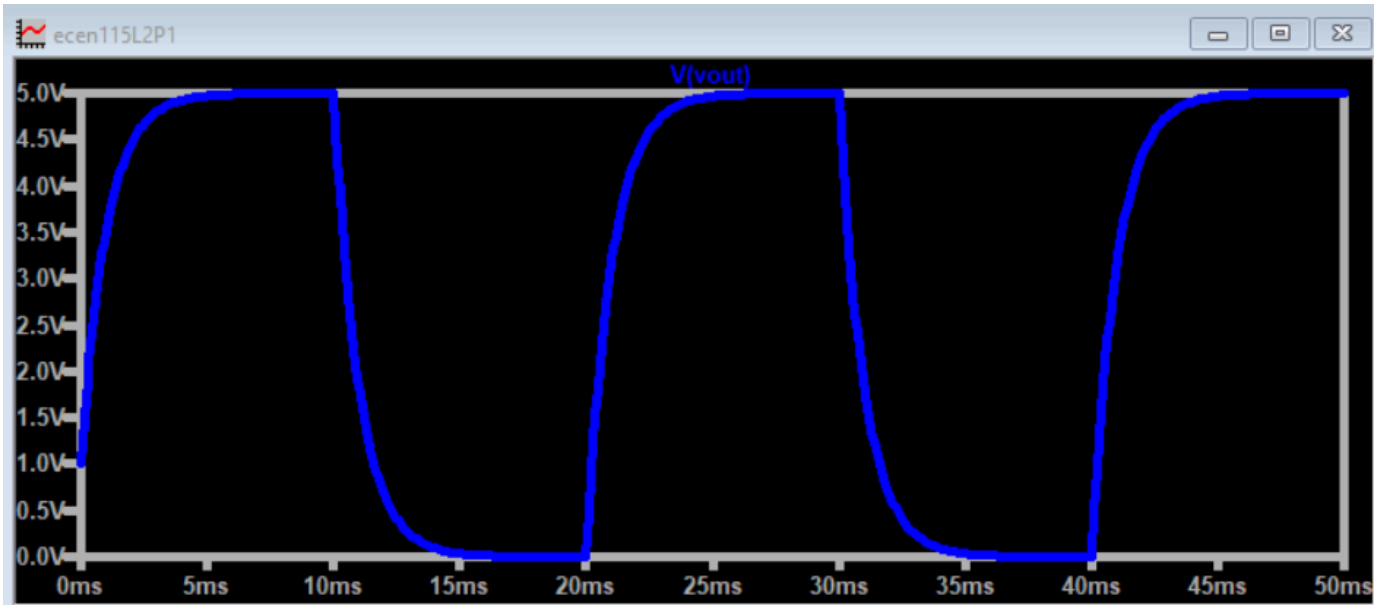
(8) Plot current i_{OUT} as a function of time.

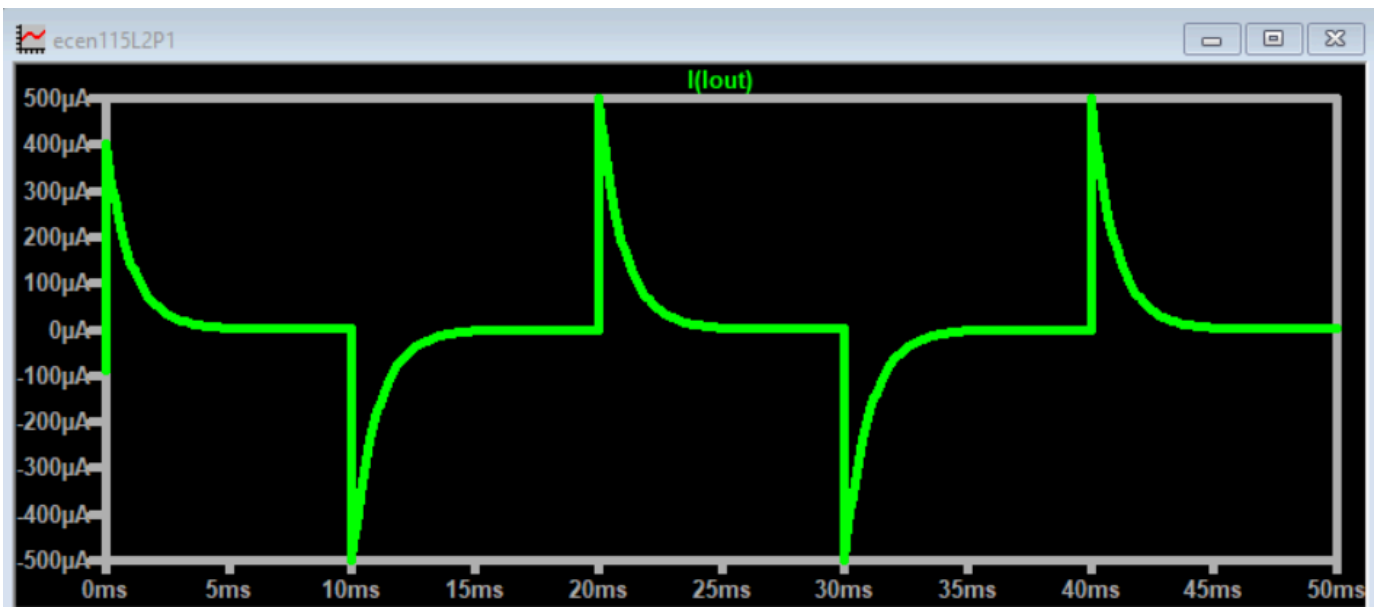


What does the plot show? Explain its shape. Verify if i_{OUT} is the same as your plot in (6)
The plots shows how the current jumps to + or - (10V/20k-Ohms) or 500microAmps and then immediately approaches 0 Amps everytime voltage changes. Iout is not the same as (6) and it shouldn't be. It is the same as (7) which is expected since C*D(v)/dt=Ic.

Make sure to take a screen capture of the simulation waveforms.

(9) Modify the initial condition on the capacitor C1 to 1V. Rerun the simulations and replot.





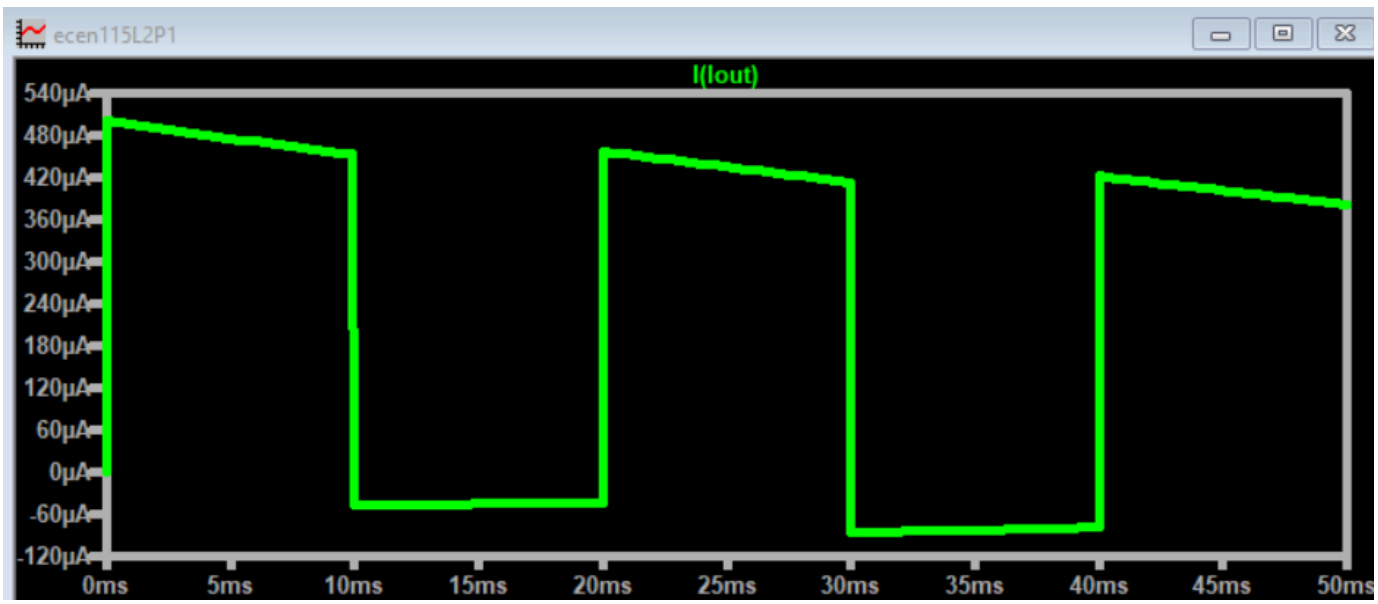
How did the waveforms change? Explain why?

For the voltage waveform, we started at 1V instead of 0V. This is due to the initial condition of V=1. Our equation to show the graphs behavior (assuming constant voltage) changes from 5-5e^-1000t to 5-4e^-1000t. So now at t-0 we get 1 instead of 0.

For the current waveform, let's take the derivative of both voltage equations to find current. This gives us 5000e^-1000t and 4000e^-1000t and now we multiply by the capacitor value .1uF. This gives us .0005e^-1000t and .0004e^-1000t. These equations are accurately displayed up until 10ms when the voltage changes, so let's just focus on the constant aspects. The initial jump part at t=0 is just the simulation. Note at t = 0, currents are .0005A and .0004A respectively.

Make sure to take a screen capture of the simulation waveforms.

(10) Modify the value of C1 = 10uF. Rerun the transient simulations and replot the signals. Note: I am getting rid of the initial condition of Vc=1.



How did the waveforms change? Explain why?

Not only do the waveforms appear to change less rapidly during the pulses, but they also exhibit a general trend up or down overall. The increased Capacitor value makes it so the capacitor itself can hold more charge. Our pulse is not long enough to fully charge this capacitor anymore. Current will just decrease proportionally to the voltage.

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Make sure to take a screen capture of the simulation waveforms.

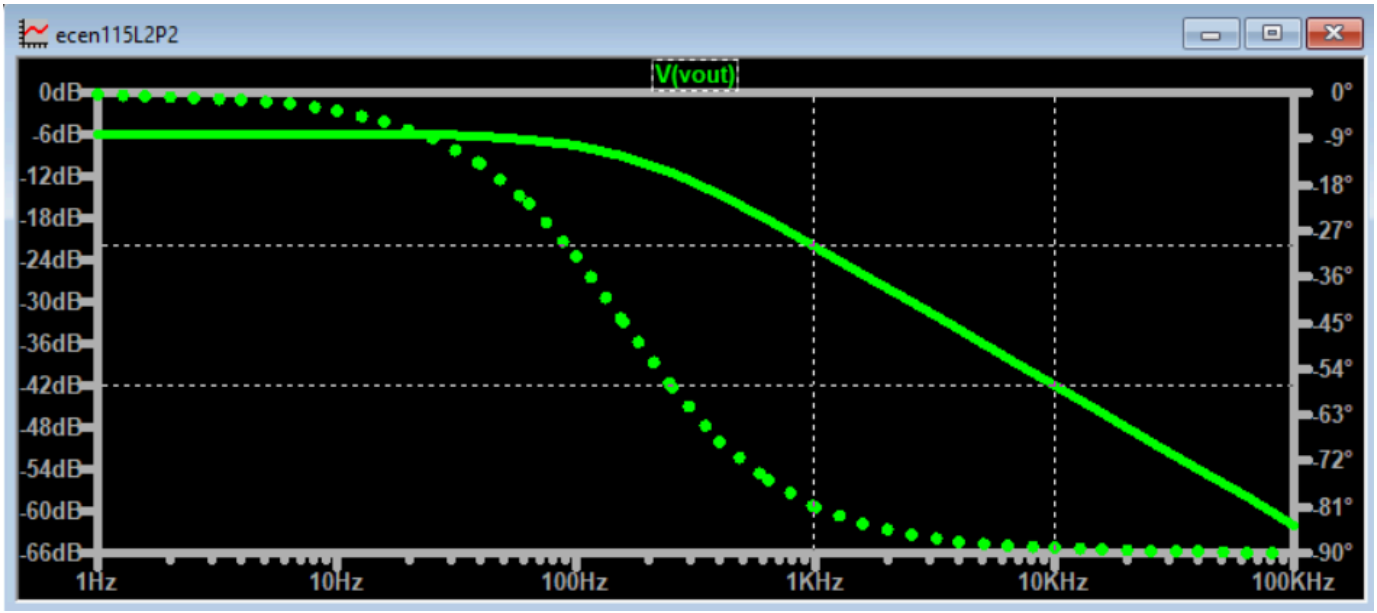
TA Check Point: Complete Part 1 and Part 2 and demo the results to your TA.

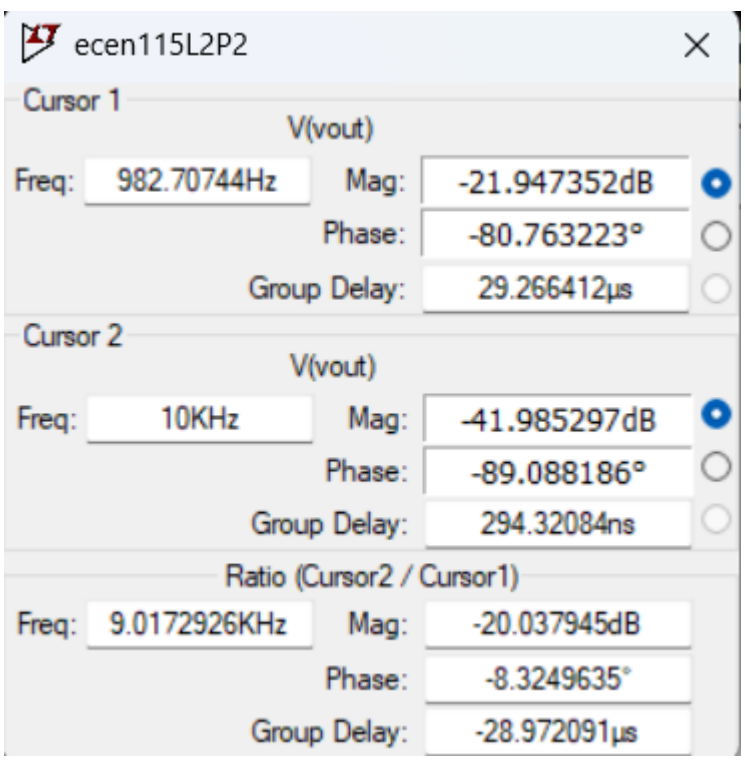
Part 3: AC analysis

(1) Assign component value for R1 = 20kΩ, R2 = 20kΩ and C1 = 0.1μF. Initial condition of 0V on the capacitor.

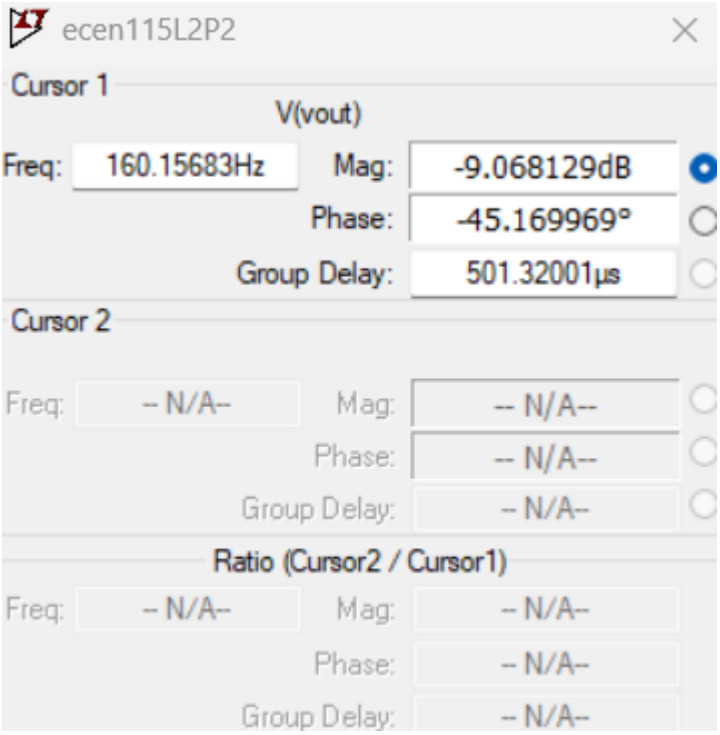
(2) V_{IN} is a sinusoidal source. Set the small signal AC analysis amplitude to 1.

(3) Simulate using AC Analysis. Sweep type: Decade.
Number of Points per decade: at least 5. Start frequency: 1Hz. Stop Frequency 100kHz.
Use cursors to measure the slope of the line to verify -20dB/decade
Measure the -3dB frequency to verify the time constant and cutoff frequency.





Baseline freq = -6db
With the -3db rule, it should be -9db at our cutoff freq of 159hz. It is.



Screen Capture the Bode plot of magnitude and phase of vout/vin.

TA Check Point: Complete Part 3 and demo the results to your TA.

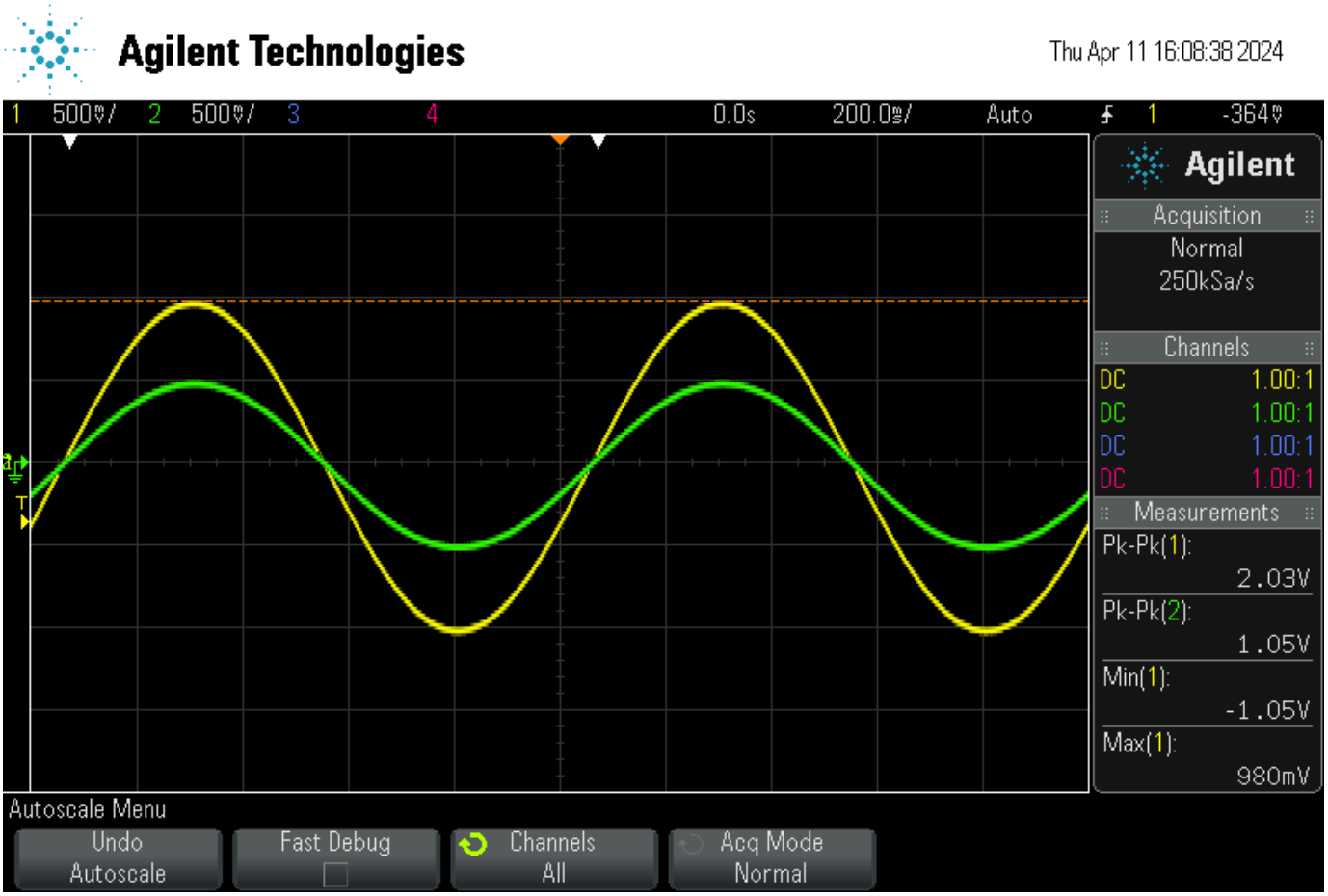
<u>Physical Circuit Build</u>

Build the circuit in Figure 1 with components R1 = 20kΩ, R2 = 20kΩ and C1 = 0.1µF. For the circuit build of Figure 1 we would like to observe its output to input of various frequencies and understand its frequency behavior. The following parts helps us make observations regarding this passive filter.

<u>Part 4: Transient input</u>

(1) Build the circuit in Figure 1 with components R1 = 20kΩ, R2 = 20kΩ and C1 = 0.1µF. (2) Connect the signal generator as the voltage source VIN with a sinusoidal input with 1V amplitude and 1Hz frequency. Connect an oscilloscope at the output to monitor the voltage. (3) Observe VIN and VOUT on the oscilloscope. Zoom in to see two to three time periods. **Do the results look as expected? Why or why not?**

Yes they do, we expect a voltage division of 2 meaning the amplitude of the output should be ½ of the input (gain of ½). We see that in this photo:



Make sure to take a screen capture of the measured waveforms.

(5) Make VIN a sinusoidal input with 1V amplitude and 10Hz frequency. Repeat step 3.

Again we see what we expect in this photo but there is a very slight phase shift:

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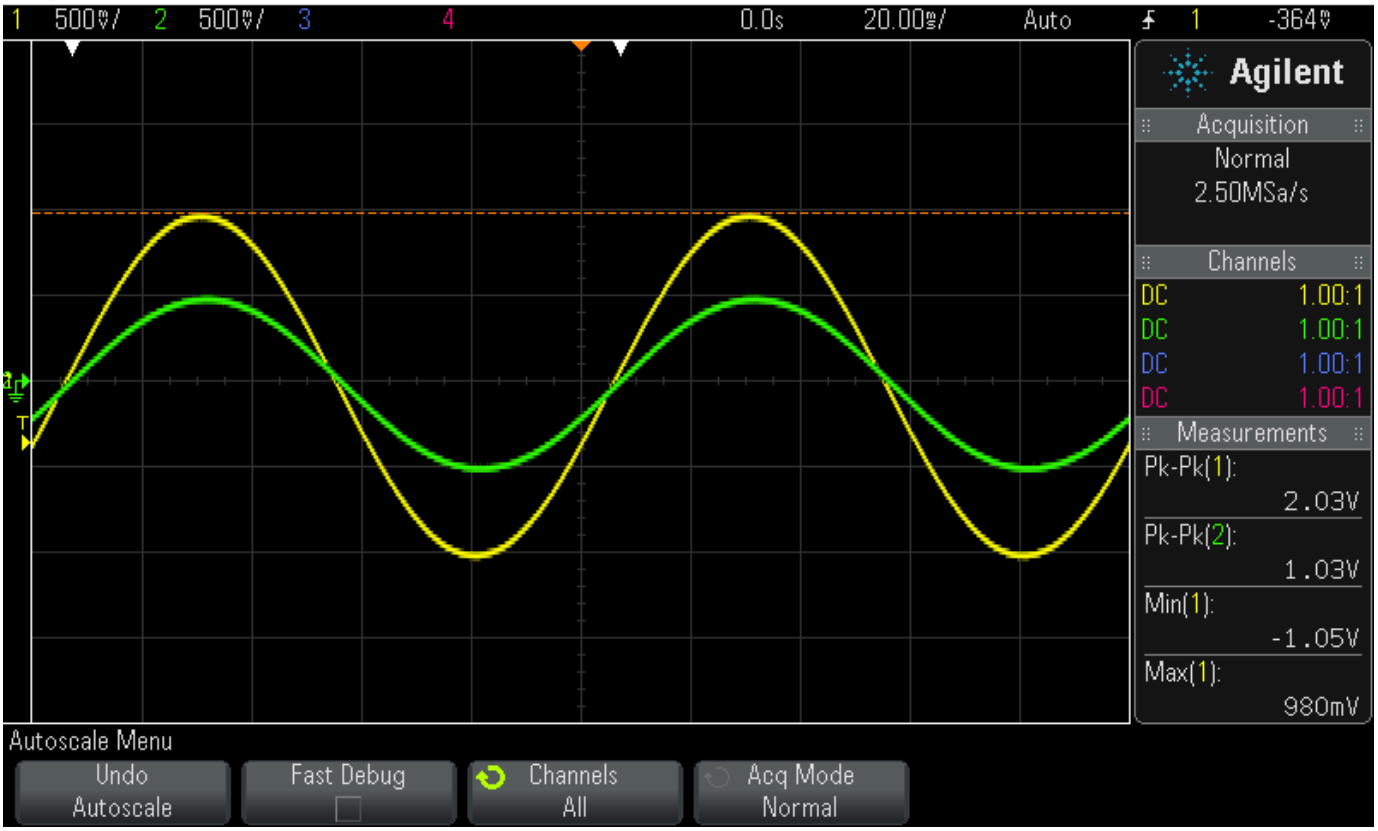
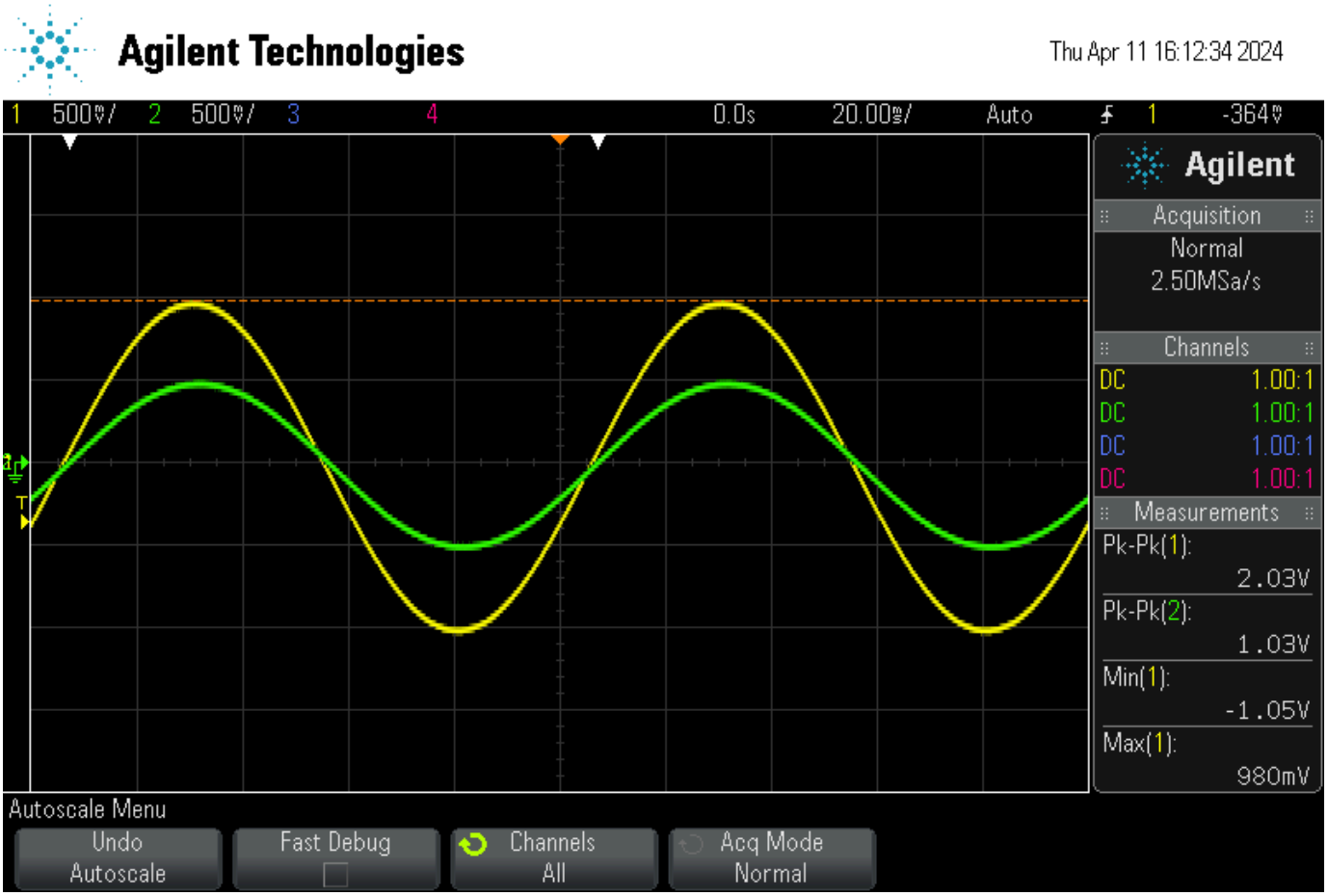
Thu Apr 11 16:12:34 2024





(6) Make v_{IN} a sinusoidal input with with 1V amplitude and 100Hz frequency. Change the transient duration to 10msec. Repeat step 3.

The ½ gain effect is still in effect, but the phase shift is much more noticeable.

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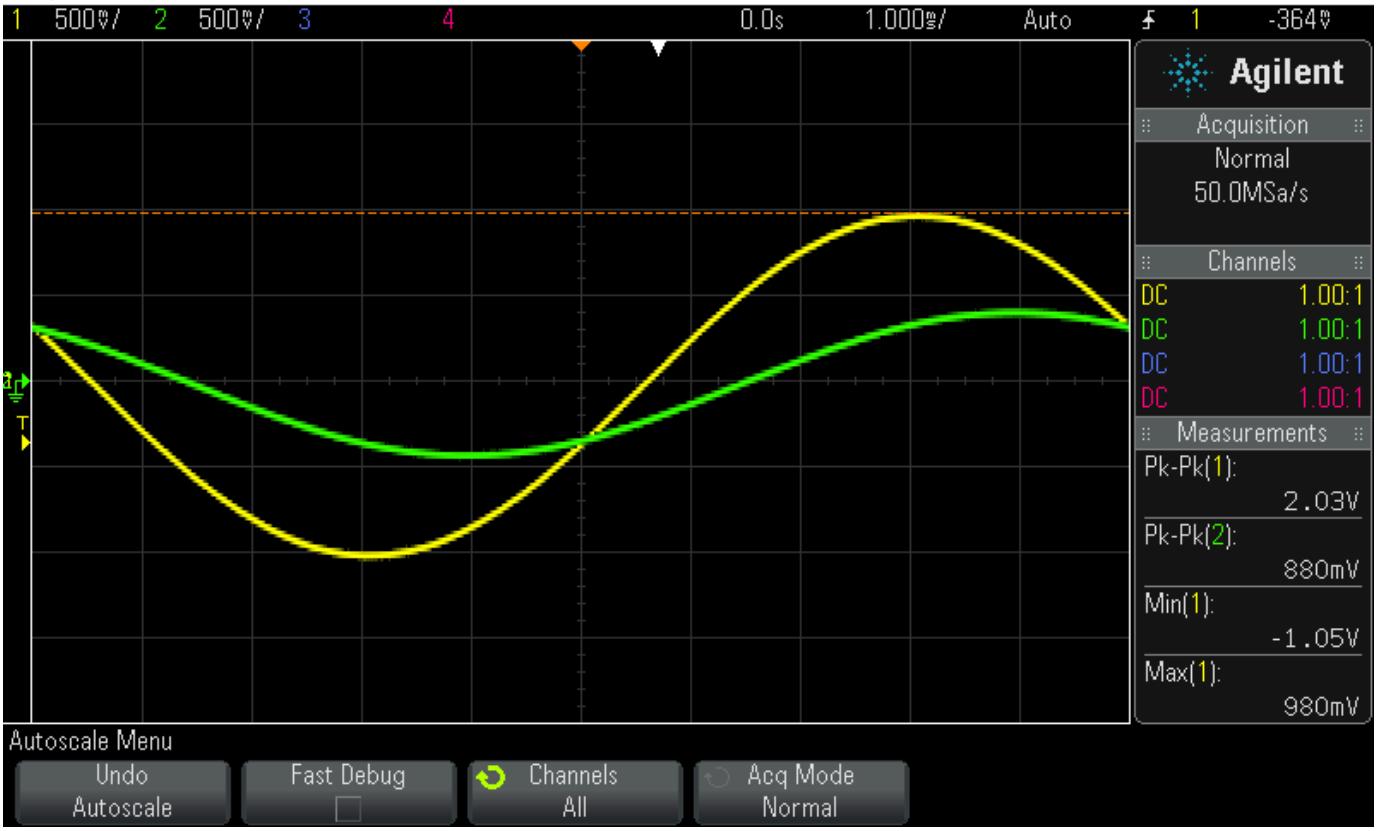

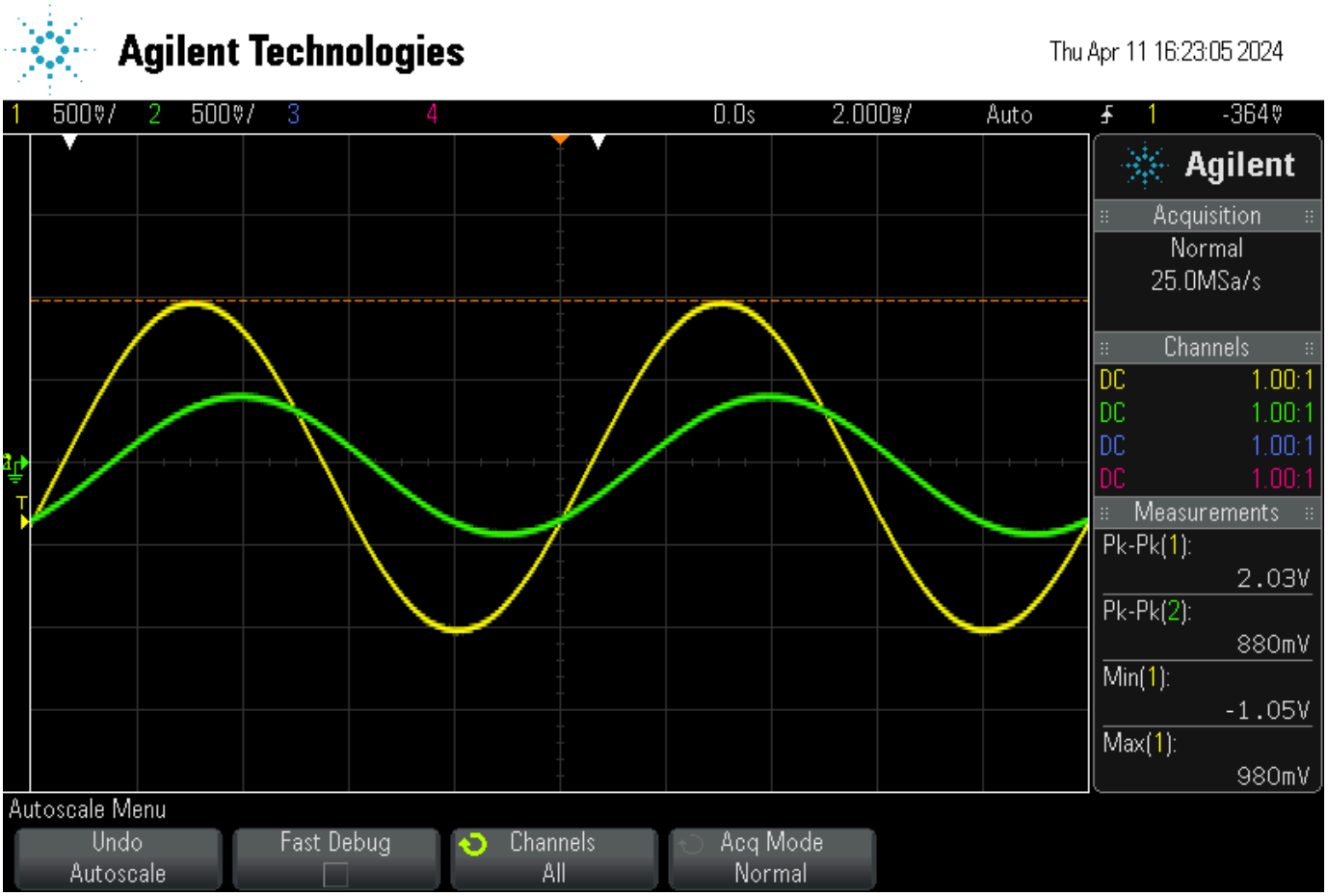

(7) Make VIN a sinusoidal input with with 1V amplitude and 1kHz frequency. Change the transient duration to 1msec. Repeat step 3.

The gain effect is gone now as 1kHz is above the cutoff freq of 159Hz. This is what we expect.

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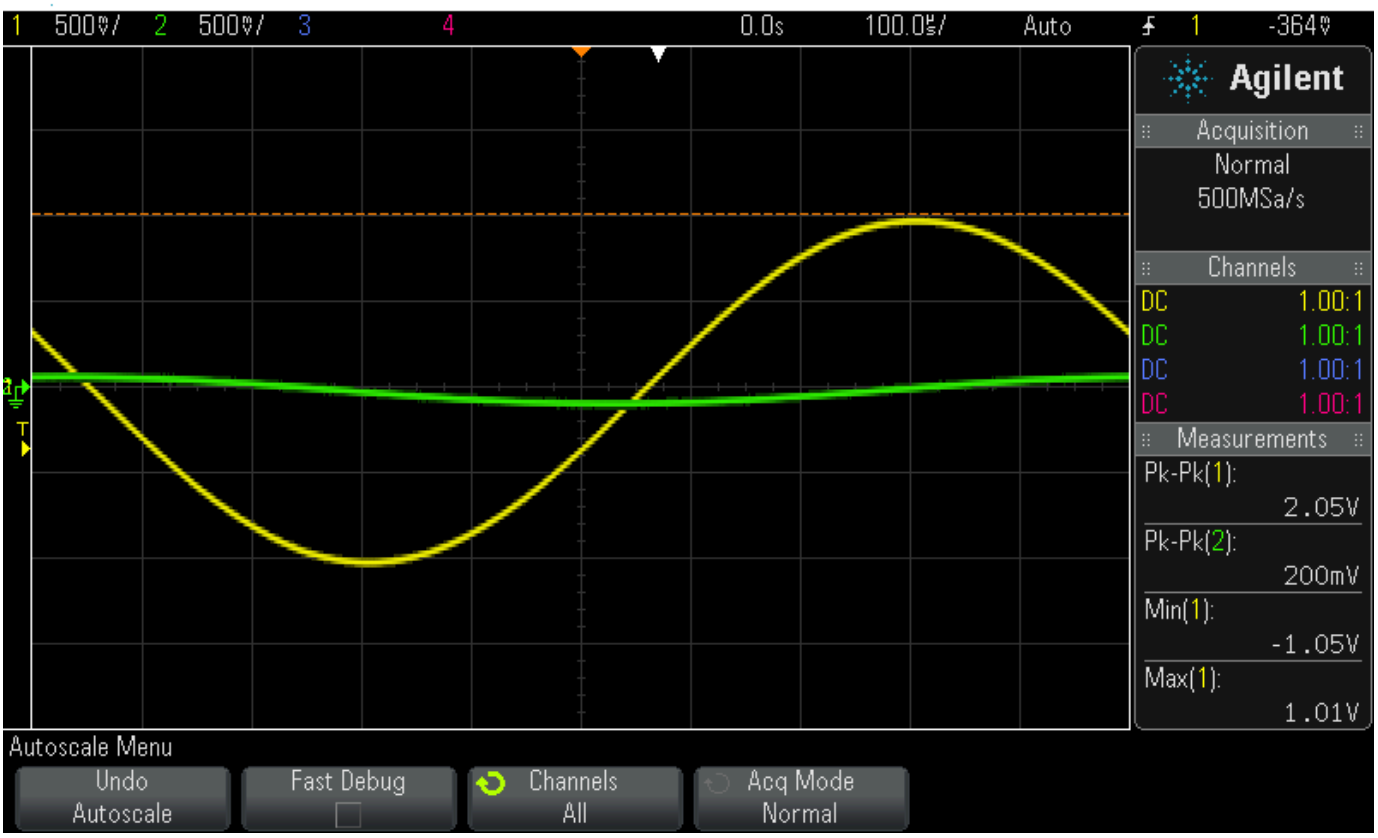
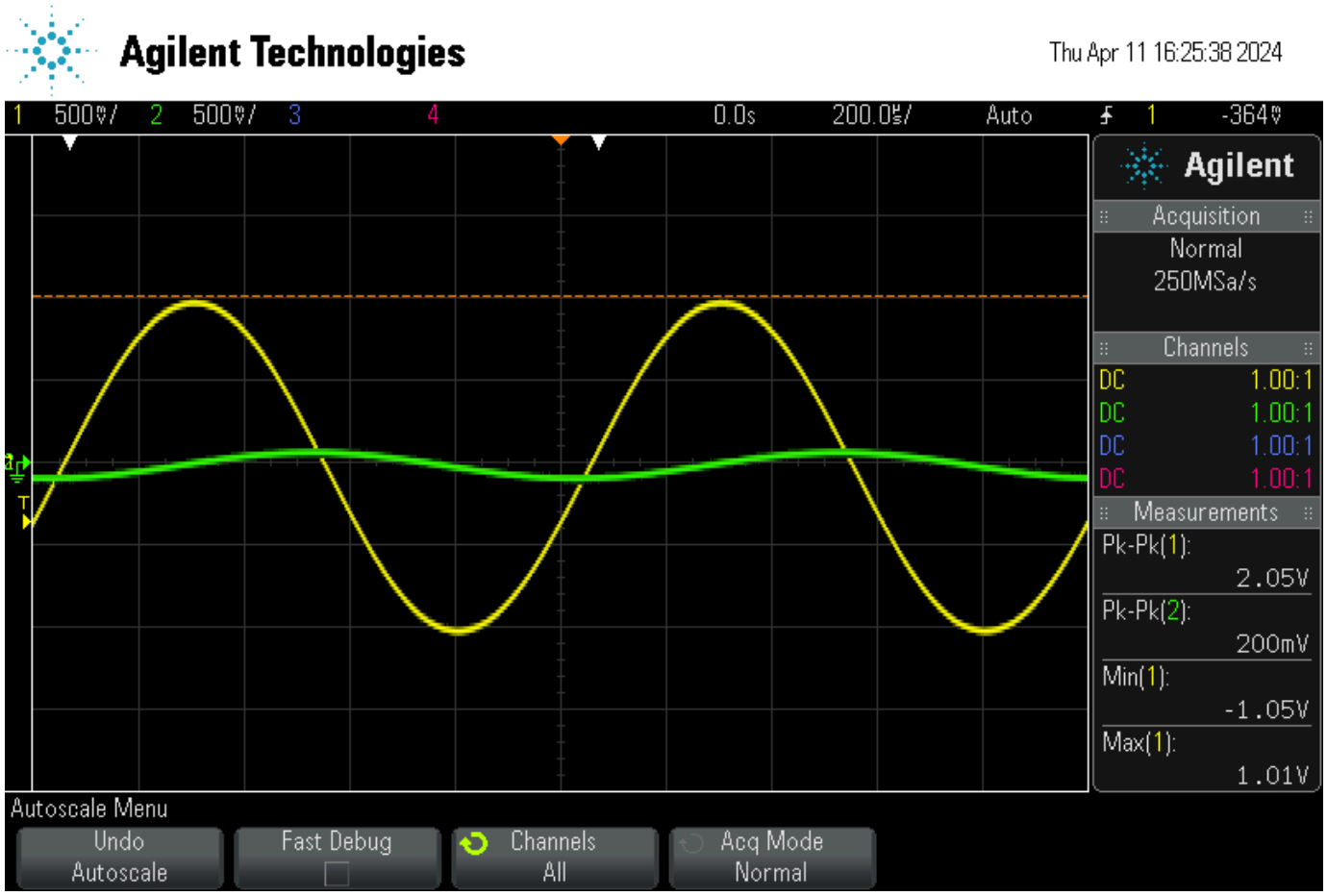
Thu Apr 11 16:25:29 2024





(8) What have you observed while doing steps (2) through (7)? What behavior does this RC circuit show?

Our circuit was intended to cutoff frequencies higher than 159hz. However, the closer we got to the cutoff frequency, even if we were still below it, the worse our phase shift got. This is because the -3db rule starts to take effect.

TA Check Point: Complete Part 4 and demo the results to your TA.

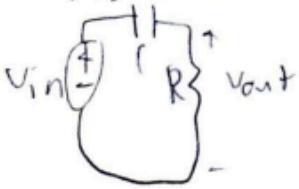
III. REPORT

Write a short laboratory report that details all the work done **and** answer the following questions:

- What is the RC time constant when R1 = 20kΩ, R2 = 20kΩ and C1 = 0.1μF?
  - The RC time constant is the following:
  - $\tau = R_{eq}C_1 = \frac{R_1R_2}{R_1+R_2}C_1 = 0.001\ s = 1\ ms$
- From part 2, what did you understand is the relationship between i_{OUT} (current in capacitor) and v_{OUT} (voltage across the capacitor)
  - We understood that I_{OUT} is directly related to the time derivative V_{OUT}, more specifically $I_{out}(t) = C_1\frac{dV_{out}(t)}{dt}$
- Draw the schematic of a single time constant high pass filter. What will be the value of the R and C to obtain a cutoff frequency 1/10th of the low pass filter you created in the lab.
- If we took the output of the high-pass-filter you just designed and made that the input of the low pass filter of the lab, what sort of behavior do we expect at the output of the combined circuit?

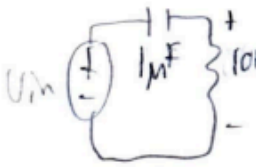
Draw a rough bode plot sketch of this behavior.
Answers to the last three questions:

High pass



$w = 2\pi f$

$\left(\frac{1}{jw10^6} + 10k\right) / V_{in} \quad \frac{10k}{\frac{1}{w10^6}+10k} \cdot \frac{\frac{1}{w10^7}}{10k+\frac{1}{w10^7}}$

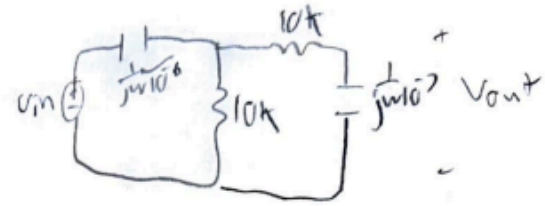


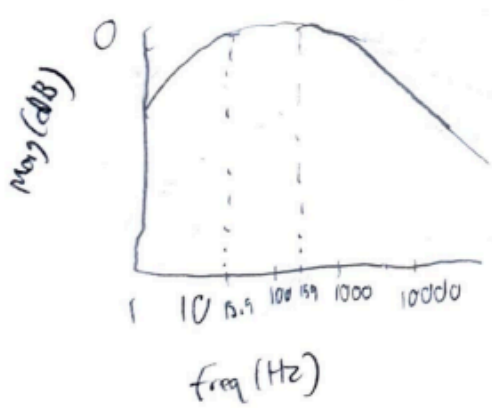
$\frac{1}{2\pi RC} = 15.915$

10% of 159 = 15.9 ✓

change C value to 1μF or Req to 100kΩ. We choose C to 1μF.

$\boxed{\begin{array}{l} R = 10k\Omega \\ C = 1\mu F \end{array}}$



$V_{in} \cdot \frac{10k}{10k+\frac{1}{jw10^6}}$

We would expect a band pass filter that blocks frequencies less than 15.9 hz and greater than 159 hz.